

Constructions of Hamiltonian graphs with prescribed parameters

Thesis

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Outline

- 1 Problem statement
- 2 Approach 1: Greedy chordal search
- 3 Approach 2: 2-lift construction
- 4 Next steps

Aim of the thesis

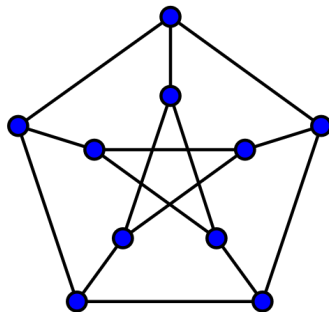
- Develop graph constructions that, for any given (k, g) -graph, produce a Hamiltonian (k, g) -graph.
- Study graphs of minimal possible order, or establish upper bounds on their order.
- Design and implement efficient algorithms for graph constructions.
- Conduct extensive computational experiments to verify graph properties.

What is a (k, g) -graph?

Definition

A (k, g) -graph is a simple graph that is:

- k -regular – every vertex has degree exactly k ,
- has *girth* g – the length of its shortest cycle is exactly g .



Example of a $(3, 5)$ -graph (Petersen)

Approach 1 – main idea

- **Goal:** construct a k -regular graph of girth $\geq g$ that remains Hamiltonian.
- **Initial structure:** start from a cycle C_n , which is already Hamiltonian.
(where n is a Moore bound)
- **Chord addition:** add edges only between vertices that still need degree, subject to the girth constraint

$$(u, v) \text{ is allowed} \iff \text{dist}_G(u, v) \geq g - 1.$$

- **Outer loop:** if k -regularity cannot be achieved, increase n (adjusted for parity if k is odd) and restart.
- **Variants:** four variants in `greedy_chordal_search/algorithms/` differ only in the rule used to select the next edge.

Approach 1 – shared greedy shell

Input: k, g

Output: k -regular Hamiltonian graph with girth $\geq g$ (if found)

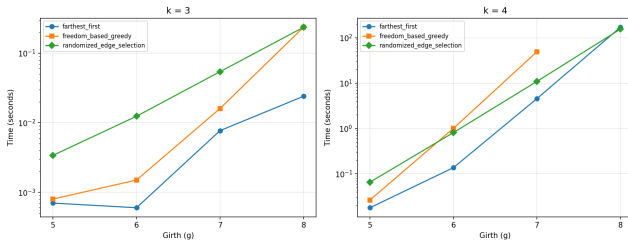
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1  $n \leftarrow n_0(k, g)$ , parity-adjusted
2 while true do
3    $G \leftarrow C_n$ 
4   while there exists a girth-legal chord between unsaturated vertices do
5      $U \leftarrow \{v : \deg_G(v) < k\}$ 
6     pick  $(u, v)$  using the active variant
7     add edge  $(u, v)$  to  $G$ 
8   end
9   if  $G$  is  $k$ -regular then return  $G$ 
10
11    $n \leftarrow n + 1$  // or  $n + 2$  if  $k$  is odd
12 end
```

Approach 1 – edge selection variants

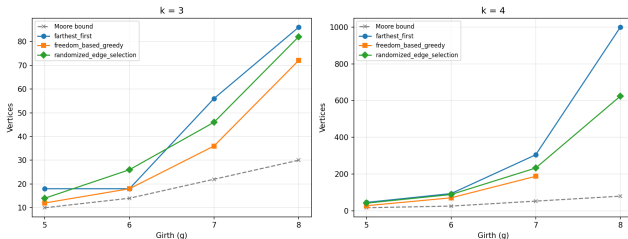
- **Randomized selection:** choose a random legal unsaturated pair.
- **Farthest-first heuristic:** connect vertices at maximum allowable distance.
- **Freedom-based greedy:** choose vertices with the fewest available options first.
- **Backtracking search:** explore candidates recursively with feasibility checks.

Approach 1 – experimental results

Execution Time vs Girth



Graph Size (Vertices) vs Girth



What is a 2-lift?

Given a graph $G = (V, E)$, the *2-lift* H has vertex set $V \cup V'$ (two disjoint copies of V). For each edge $uv \in E$, assign a sign $\sigma(uv) \in \{0, 1\}$:

- $\sigma(uv) = 0$ (parallel): edges uv and $u'v'$,
- $\sigma(uv) = 1$ (cross): edges uv' and $u'v$.

Properties:

- $|V(H)| = 2|V(G)|$, and regularity is preserved.
- Girth is often preserved or increased depending on σ .
- Edge choices control cycle structure.

Basic construction

Input: Connected k -regular graph G

Output: 2-lift H

- 1 Compute a 2-factor decomposition of G
 - 2 Initialize all $\sigma(e) = 0$
 - 3 Select one edge per cycle and set $\sigma(e) = 1$
 - 4 Construct a meta-graph of cycles
 - 5 Compute a spanning tree of the meta-graph
 - 6 Set corresponding edges to cross edges
 - 7 **return** resulting 2-lift H
-

- For experimental testing, we used the graph database *House of Graphs*.
- Hamiltonicity checking was performed using standard computational resources, without relying on high-performance computing infrastructure (e.g., supercomputers).
- The available computational resources were sufficient for all tested graph instances.

Issue with the basic construction

Testing graphs from the *House of Graphs* database showed that the basic construction does not always produce a Hamiltonian 2-lift.

Main difficulties:

- multiple possible 2-factor decompositions,
- multiple valid spanning trees of the meta-graph,
- sensitivity of Hamiltonicity to edge selection.

More 2-factors imply a greater number of spanning trees, and consequently provide more choices for selecting cross edges.

Therefore, we tested all spanning trees for non-Hamiltonian cases.

Empirical result

Across 27 test graphs (3-regular, connected, fixed girth):

- Exhaustive spanning-tree search found a Hamiltonian 2-lift for every graph.

Next steps

- Formally characterize the optimal spanning-tree selection strategy.
- Improve constructions to achieve smaller graph orders.

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Thank you.